

Heat & Thermodynamics

Thermodynamic Potentials: In thermodynamic, thermodynamic potentials are the parameters having the dimension of energy and describe the amount of potential energy in a thermodynamic system under some constraints. The different potentials correspond to different constraints to which the system may be subjected. The four most common thermodynamic potentials are:

- Internal Energy
- Helmholtz Free Energy
- Enthalpy
- Gibbs Free Energy

Internal Energy:

In thermodynamics, the internal energy of a thermodynamic system with well-defined boundaries, denoted by U , is the total kinetic energy due to the motion of molecules (translational, rotational, vibrational) and the potential energy associated with the vibrational electric energy of atoms within molecules or crystals. The internal energy cannot be precisely measured, only the change in the internal energy can be measured, and the total internal energy of a given system is the difference between the internal energy of the system and the internal energy of the same system at absolute zero temperature. Since, the absolute zero cannot be attained, the total internal energy cannot be precisely measured.

From the first law of thermodynamic,

$$dQ = dU + dW$$

$$\text{Or, } dU = dQ - PdV$$

$$\text{Where, } dW = PdV$$

$$dU = TdS - PdV$$

$$\text{Where, } dQ = TdS$$

Helmholtz Free Energy:

The Helmholtz free energy is a thermodynamic potential, which measures the useful work obtainable from constant temperature and constant volume thermodynamic system. It is sometimes known as “work Constant”. For a simple system, with a fixed number of particles, the negative of the difference in the Helmholtz free energy is equal to the maximum amount of work extractable from a thermodynamic process in which temperature is held constant. It is defined as:

$$F = U - TS$$

Where,

F = Helmholtz Free Energy (Unit: Joule)

U = Internal Energy of the system (Unit: Joule)

T = Absolute Temperature (Unit: Kelvin)

S = Entropy (Unit: Joule/Kelvin)

Enthalpy:

Enthalpy is the sum of the internal energy of matter and the product of its volume and pressure. It is also called heat constant. It is mathematically defined as:

$$H = U + PV$$

Where,

H = Enthalpy (Unit: Joule)

U = Internal Energy of the system (Unit: Joule)

P = Pressure (Unit: Pascal)

V = Volume (Unit: Cubic meter)

Gibbs Free Energy:

In thermodynamics, the Gibbs free energy is a thermodynamic potential and is therefore a state function of a thermodynamic system.

Mathematically it is defined as:

$$G = U + PV - TS$$

Where,

G = Gibbs Free Energy (Unit: Joule)

U = Internal Energy of the system (Unit: Joule)

P = Pressure (Unit: Pascal)

V = Volume (Unit: Cubic meter)

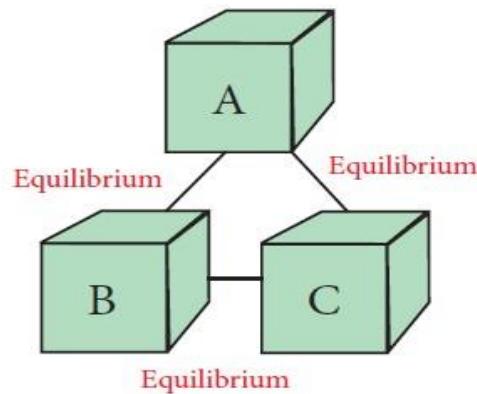
T = Absolute Temperature (Unit: Kelvin)

S = Entropy (Unit: Joule/Kelvin)

The Gibbs free energy is one of the most important thermodynamic functions for characterization of a system. It is a factor in determining outcomes such as the voltage of an electrochemical cell and the equilibrium constant for a reversible reaction. The negative value of the change in Gibbs free energy ΔG means that such a reaction will be favored and release energy in the form of work. Conversely, if conditions indicated a positive ΔG then energy, in the form of work, would have to be added to the reacting system to make the reaction go.

The Zeroth Law of Thermodynamics:

The zeroth law of thermodynamics states that if two systems are both in thermal equilibrium with a third system, then the first two systems are also in thermal equilibrium with each other.

**Third Law of Thermodynamics:**

In all heat energies, there is always loss of heat in the form of conduction, radiation and friction.

Therefore, in actual heat energies $\frac{Q_1}{T_1}$ is not equal to $\frac{Q_2}{T_2}$. So, $\frac{Q_1}{T_1} - \frac{Q_2}{T_2}$ is not zero but a positive quantity.

When cycle after cycle is repeated, the entropy of the system increases and tends to a maximum value. With the increase in entropy, the disorder of the molecules of a substance increases. With the decrease in entropy, the disorder decreases. Generalizing from experience, we may state that it is not possible to attain 100% orderliness. “Hence by no finite series of process the absolute zero is attainable “. This is known as “Principle of the unattainability of absolute zero”.